

LEXAN™ 945 resin

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General Information

Product Description

Lexan® 945 Polycarbonate (PC) resin is a non-filled, injection moldable grade. This non-chlorinated, non-brominated flame retardant PC has an UL-94 V0 rating and is available in various opaque color options. Lexan 945 is a general purpose resin designed to meet the needs of various applications.

General

Material Status	• Commercial: Active
Availability	• North America
Additive	• Flame Retardant
Features	• Bromine Free • Flame Retardant • Chlorine Free • General Purpose
Uses	• General Purpose
Appearance	• Colors Available • Opaque
Processing Method	• Injection Molding
Multi-Point Data	• Coefficient of Thermal Expansion vs. Temperature (ASTM E831) • Flexural DMA (ASTM D4065) • Pressure-Volume-Temperature (PVT - Zoller Method) • Shear DMA (ASTM D4065) • Specific Heat vs. Temperature (ASTM D3417) • Tensile Fatigue • Tensile Stress vs. Strain (ASTM D638) • Thermal Conductivity vs. Temperature (ASTM E1530) • Viscosity vs. Shear Rate (ASTM D3835)

ASTM & ISO Properties ¹

Physical	Nominal Value	Unit	Test Method
Specific Gravity	1.19		ASTM D792
Melt Mass-Flow Rate (MFR) (300°C/1.2 kg)	10	g/10 min	ASTM D1238
Molding Shrinkage - Flow (0.126 in)	0.0060 to 0.0080	in/in	Internal Method
Mechanical	Nominal Value	Unit	Test Method
Tensile Modulus ²	330000	psi	ASTM D638
Tensile Strength ³ (Yield)	9000	psi	ASTM D638
Tensile Strength ³ (Break)	9500	psi	ASTM D638
Tensile Elongation ³ (Yield)	6.0	%	ASTM D638
Tensile Elongation ³ (Break)	130	%	ASTM D638
Flexural Modulus ⁴ (1.97 in Span)	340000	psi	ASTM D790
Flexural Strength ⁴ (Yield, 1.97 in Span)	14700	psi	ASTM D790

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Impact	Nominal Value	Unit	Test Method
Notched Izod Impact (73°F)	15	ft-lb/in	ASTM D256
Instrumented Dart Impact (73°F, Total Energy)	650	in-lb	ASTM D3763
Thermal	Nominal Value	Unit	Test Method
Deflection Temperature Under Load 66 psi, Unannealed, 0.126 in	280	°F	ASTM D648
Deflection Temperature Under Load 264 psi, Unannealed, 0.126 in	260	°F	ASTM D648
Vicat Softening Temperature	290	°F	ASTM D1525 ⁵
CLTE - Flow (-40 to 104°F)	0.000037	in/in/°F	ASTM E831
CLTE - Transverse (-40 to 104°F)	0.000037	in/in/°F	ASTM E831
RTI Elec	266	°F	UL 746
RTI Imp	248	°F	UL 746
RTI Str	266	°F	UL 746
Electrical	Nominal Value	Unit	Test Method
Arc Resistance ⁶	PLC 7		ASTM D495
Comparative Tracking Index (CTI)	PLC 2		UL 746
High Amp Arc Ignition (HAI)	PLC 3		UL 746
High Voltage Arc Tracking Rate (HVTR)	PLC 4		UL 746
Hot-wire Ignition (HWI)	PLC 2		UL 746
Flammability	Nominal Value	Unit	Test Method
Flame Rating			UL 94
0.0315 in	V-2		
0.0450 in	V-0		
0.120 in	5VA		
Glow Wire Flammability Index (0.0394 in)	1760	°F	IEC 60695-2-12
Glow Wire Ignition Temperature (0.0394 in)	1610	°F	IEC 60695-2-13
Oxygen Index	35	%	ISO 4589-2

Processing Information

Injection	Nominal Value	Unit
Drying Temperature	250	°F
Drying Time	3.0 to 4.0	hr
Drying Time, Maximum	48	hr
Suggested Max Moisture	0.020	%
Suggested Shot Size	40 to 60	%
Rear Temperature	520 to 560	°F
Middle Temperature	540 to 580	°F
Front Temperature	560 to 600	°F
Nozzle Temperature	550 to 590	°F
Processing (Melt) Temp	560 to 600	°F
Mold Temperature	160 to 200	°F
Back Pressure	50.0 to 100	psi

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Injection	Nominal Value	Unit
Screw Speed	40 to 70	rpm
Vent Depth	0.0010 to 0.0030	in

Notes

¹ Typical properties: these are not to be construed as specifications.

² 2.0 in/min

³ Type I, 2.0 in/min

⁴ 0.051 in/min

⁵ Rate B (120°C/h), Loading 2 (50 N)

⁶ Tungsten Electrode

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